



ML-based wireless power transfer

Researchers at Chiba University (Chiba, Japan) have developed a machine learning-based method to design wireless power transfer (WPT) systems that stay efficient and stable across varying loads. Unlike traditional models that rely on



Researchers developed a fully numerical design method using differential equations and genetic algorithms to optimise WPT systems. This approach ensures stable output voltage, high efficiency, and zero-voltage switching across varying loads, overcoming the limitations of traditional analytical-based methods (Credit: Wikimedia Commons)

idealised equations, the team used differential equations and a genetic algorithm to optimise real-world performance. Their system achieved 86.7% efficiency at 6.78MHz with under 5% voltage fluctuation, paving the way for compact, self-optimising, load-independent WPT solutions.

Low-power, reliable transmitter chip

Researchers at MIT (United States) have developed a compact transmitter chip that reduces signal errors by a factor of four and extends battery life for IoT devices. Its secret: an energy-saving modulation scheme that adapts to wireless conditions while adding smart padding bits for error-free decoding. Powered by MIT's GRAND algorithm, the chip ensures consistent, low-power transmissions, even in noisy settings, making it ideal for 6G, industrial monitoring, and smart home devices.

Real-time, private AI on mobile, wearable, and edge systems

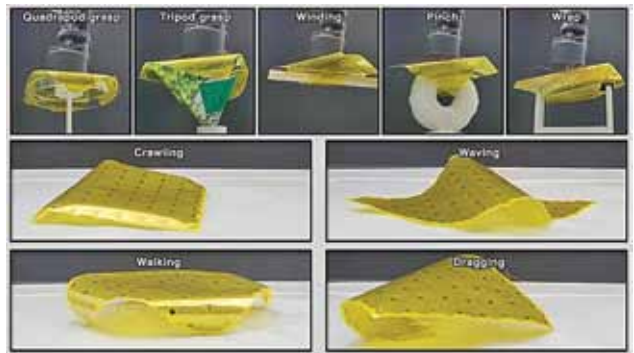
Sejong University's (Seoul, South Korea) STAU hardware brings large language models like BERT and GPT to embedded devices, removing the need for cloud-based processing. Achieving up to 5.18x speedup and 97%+ accuracy, STAU uses a variable systolic array, radix-2 softmax, and custom 16-bit floating point to boost efficiency. It enables real-time, private AI on mobile, wearable, and edge systems.

Ultra-low-voltage white OLED

Tokyo scientists have developed a white OLED that operates at a voltage of less than 1.5 volts, far lower than the typical 2.5 volts. Using a triplet-triplet annihilation (TTA) mechanism, the team enabled efficient blue light at reduced energy cost. Doping with sky-blue and yellow emitters yielded balanced white light. This ultra-low-voltage design promises battery-powered displays for wearables, lighting, and mobile devices, ushering in a new era of energy-efficient OLEDs.

Origami-inspired robotic sheet

For adaptive robotics, KAIST (University in Daejeon, South Korea) researchers have created an origami-inspired robotic sheet that folds, crawls, and grasps while sensing its envi-

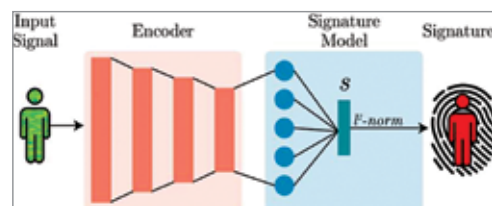


Shape shifting robotic sheet (Credit: Nature Communications, 2025)

ronment in real time. Embedded with 308 heater-sensors, it achieves precise, programmable shape-shifting from -87° to 109° . Potential applications span exploration, haptics, and healthcare, enabling flexible, intelligent operation where conventional robots fall short.

Wi-Fi that knows who you are

WhoFi, developed at La Sapienza University (Rome, Italy), is a Wi-Fi-based surveillance system that identifies individuals by how their bodies disrupt wireless signals; no cameras, contact, or consent is needed. Using Channel State Information and AI, it achieves 95.5% accuracy, even through walls or in darkness. While promising for security, WhoFi



Overview of the proposed framework (Credit: arXiv, 2025)

raises serious privacy concerns as it passively tracks movement without detection.